

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Name</b>         | <b><u>Dichloroisocyanuric acid sodium salt</u></b>                                                                                  |
| <b>CAS No</b>               | 2893-78-9                                                                                                                           |
| <b>Synonyms</b>             | 1-Sodium-3,5-dichloro-s-triazine-2,4,6-trione; Dichloro-s-triazine-2,4,6-(1H,3H,5H)-trione sodium salt; Sodium dichloroisocyanurate |
| <b>Molecular Formula</b>    | C3 Cl2 N3 Na O3                                                                                                                     |
| <b>Molecular Weight</b>     | 219.95                                                                                                                              |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                                                                               |
| <b>Uses advised against</b> | No Information available                                                                                                            |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>B23504</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number**     **HSR001324**

### GHS Classification

#### Physical hazards

Oxidizing solids Category 2

#### Health hazards

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

**Label Elements**



**Signal Word**

**Danger**

**Hazard Statements**

H302 - Harmful if swallowed  
H319 - Causes serious eye irritation  
H272 - May intensify fire; oxidizer  
H335 - May cause respiratory irritation  
H410 - Very toxic to aquatic life with long lasting effects  
H315 - Causes skin irritation

**Precautionary Statements**

**Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P220 - Keep away from clothing and other combustible materials  
P221 - Take any precaution to avoid mixing with combustibles  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P273 - Avoid release to the environment

**Response**

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P391 - Collect spillage

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Lachrymator (substance which increases the flow of tears) Toxicity to Soil Dwelling Organisms  
Toxic to terrestrial vertebrates  
Contact with acids liberates toxic gas

## Section 3 - Composition and Information on Ingredients

| Component                             | CAS No    | Weight % |
|---------------------------------------|-----------|----------|
| Dichloroisocyanuric acid, sodium salt | 2893-78-9 | >95      |

## Section 4 - First Aid Measures

**Description of first aid measures**

|                                            |                                                                                                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance. Do not breathe dust/fume/gas/mist/vapors/spray. Do not get in eyes, on skin, or on clothing. |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                   |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                                                                                   |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                              |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                                                                    |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Get medical attention.                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                               |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                         |
| <b>Most important symptoms and effects</b> | No information available.                                                                                                                                                                    |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                       |

## **Section 5 - Fire Fighting Measures**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

May ignite combustibles (wood, paper, oil, clothing, etc.). Risk of explosion by shock, friction, fire or other sources of ignition. Do not allow run-off from fire-fighting to enter drains or water courses. Oxidizer: Contact with combustible/organic material may cause fire.

**Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>).

**Decomposition Temperature**

230 °C

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **Section 6 - Accidental Release Measures**

**Personal Precautions, Protective Equipment and Emergency Procedures**

**Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep away from clothing and other combustible materials. Avoid dust formation.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

**Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from clothing and other combustible materials. Avoid dust formation. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas).

**Hygiene Measures**

When using do not eat, drink or smoke. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Provide regular cleaning of equipment, work area and clothing.

**Conditions for Safe Storage, Including any Incompatibilities**

**Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials.

**Incompatible Materials**

Strong oxidizing agents. Strong bases. Acids.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters**

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material            | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|---------------------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Neoprene, | See manufacturers | -               |                 | (minimum requirement) |

|                                      |             |
|--------------------------------------|-------------|
| Natural rubber, PVC. recommendations | AS/NZS 2161 |
|--------------------------------------|-------------|

Inspect gloves before use.  
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)  
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.  
Remove gloves with care avoiding skin contamination.

|                                                                   |                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>                                   | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                    |
| <b>Respiratory Protection</b>                                     | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b><br><b>Recommended half mask:-</b> | Particulates filter conforming to EN 143 (or AUS/NZ equivalent)<br>Particle filtering: EN149:2001 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                      |
| <b>Hygiene Measures</b>                                           | When using do not eat, drink or smoke. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Provide regular cleaning of equipment, work area and clothing.                                                                                                                                |
| <b>Environmental exposure controls</b>                            | Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid                    |                                          |
| <b>Appearance</b>                              | White                    |                                          |
| <b>Odor</b>                                    | Slight chlorine          |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | 6.0-7.0                  | 1% aq.sol. 25°C                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | 230 °C                   |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Water Solubility</b>                        | 30g/100ml (25°C)         |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Particle characteristics</b>                | No data available        |                                          |
| <b>Other information</b>                       |                          |                                          |
| <b>Molecular Formula</b>                       | C3 Cl2 N3 Na O3          |                                          |
| <b>Molecular Weight</b>                        | 219.95                   |                                          |
| <b>Evaporation Rate</b>                        | Not applicable - Solid   |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Yes                                                                                                                                                                                                                                   |
| <b>Stability</b>                        | Oxidizer: Contact with combustible/organic material may cause fire. Moisture sensitive. Risk of explosion by shock, friction, fire or other sources of ignition.                                                                      |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                                              |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                                                                              |
| <b>Hazardous Reactions</b>              | Contact with acids liberates toxic gas.                                                                                                                                                                                               |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Combustible material, Avoid dust formation, Exposure to moist air or water, Ignitions sources - heat, sparks and open flames, Do not subject to grinding/shock/friction, Heat, flames and sparks. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong bases, Acids.                                                                                                                                                                                         |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Nitrogen oxides (NO <sub>x</sub> ).                                                                                                                                          |

## Section 11 - Toxicological Information

### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Irritating to respiratory system. May be harmful if inhaled.                                          |
| <b>Eyes</b>       | Irritating to eyes. Lachrymator (substance which increases the flow of tears).                        |
| <b>Skin</b>       | Irritating to skin. May be harmful in contact with skin.                                              |
| <b>Ingestion</b>  | Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |

#### Numerical measures of toxicity

##### (a) acute toxicity;

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Category 4                                                       |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met |
| <b>Inhalation</b> | Based on available data, the classification criteria are not met |

| Component                             | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation                   |
|---------------------------------------|---------------------------|---------------------------|-----------------------------------|
| Dichloroisocyanuric acid, sodium salt | LD50 = 1823 mg/kg ( Rat ) | LD50 > 5000 mg/kg ( Rat ) | LC50 0.27 - 1.17 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| (e) germ cell mutagenicity; | No data available                                                              |
| (f) carcinogenicity;        | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | No data available                                                              |
| (h) STOT-single exposure;   | Category 3                                                                     |
| Results / Target organs     | Respiratory system                                                             |
| (i) STOT-repeated exposure; | No data available                                                              |
| Target Organs               | No information available.                                                      |
| (j) aspiration hazard;      | Not applicable<br>Solid                                                        |

**Symptoms / effects, both acute and delayed**

No information available.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity**

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                             | Freshwater Fish                                                                                                                                                                                                                                                                                           | Water Flea                                                                                        | Freshwater Algae | Microtox |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------|----------|
| Dichloroisocyanuric acid, sodium salt | LC50: 0.25 - 1 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.207 - 0.389 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: = 0.29 mg/L, 96h (Oncorhynchus mykiss)<br>LC50: 0.176 - 0.267 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.13 - 0.36 mg/L, 96h static (Oncorhynchus mykiss) | EC50: 0.093 - 0.16 mg/L, 48h (Daphnia magna)<br>EC50: 0.00018 - 0.00021 mg/L, 48h (Daphnia magna) |                  |          |

**Terrestrial ecotoxicity**

There is no data for this product

**Persistence and Degradability****Persistence**

Soluble in water, Persistence is unlikely, based on information available.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

**Mobility** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

**Other adverse effects**

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## **Section 13 - Disposal Considerations**

**Waste treatment methods**

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## **Section 14 - Transport Information**

**NZS 5433:2020**

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2465                        |
| <b>Proper Shipping Name</b> | DICHLOROISOCYANURIC ACID, DRY |
| <b>Hazard Class</b>         | 5.1                           |
| <b>Packing Group</b>        | II                            |

**IATA**

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2465                        |
| <b>Proper Shipping Name</b> | DICHLOROISOCYANURIC ACID, DRY |
| <b>Hazard Class</b>         | 5.1                           |
| <b>Packing Group</b>        | II                            |

**IMDG/IMO**

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2465                        |
| <b>Proper Shipping Name</b> | DICHLOROISOCYANURIC ACID, DRY |
| <b>Hazard Class</b>         | 5.1                           |
| <b>Packing Group</b>        | II                            |

**Environmental hazards**

Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable, packaged goods

**Special Precautions**

No special precautions required. Please refer to the applicable dangerous goods



regulations for additional information.

**Additional information**

None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR001324 |
|-----------------------------|-----------|

**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions according to EU REACH**

| Component                             | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dichloroisocyanuric acid, sodium salt | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                             | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|---------------------------------------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Dichloroisocyanuric acid, sodium salt | 2893-78-9 | X     | X    | 220-767-7 | -      | -   | KE-10215 | X     | X    |

| Component                             | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|---------------------------------------|-----------|------|-----------------------------------------------|-----|-----|-------|------|------|
| Dichloroisocyanuric acid, sodium salt | 2893-78-9 | X    | ACTIVE                                        | X   | -   | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDSL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                     | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  |                                                                                                                              |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Chemical incident response training.

|                         |                |
|-------------------------|----------------|
| <b>Revision Date</b>    | 15-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet